

Name: Francisco Morales Figueroa **U. Galway ID:** 25247209

Name: Julien Jessmon **U. Galway ID:** 25239966

```
%pip install matplotlib numpy scikit-learn pandas --quiet
```

```
# Package imports
import matplotlib.pyplot as plt
import sklearn
import pandas as pd
import numpy as np
```

✓ Task 1 Implement Logistic Regression

Requirements

- Neural Network Logistic regression
- Handle different number of input features
- Handle different input training samples

```
# Prediction function
def predict(Xi, W, b):
    """ This function computes the predicted probabilities for a single input
    sample Xi using the parameters W and b. It applies sigmoid activation function.
    """
    # Linear transformation
    z = np.dot(Xi, W) + b
    # Sigmoid activation function
    y_hat = 1 / (1 + np.exp(-z))
    return y_hat
```

```
# Stochastic Gradient Descent function
def stochastic_gradient_descent(X, y, W, b, alpha=0.01, max_iters=1000, threshold=1e-4):
    """ This function performs stochastic gradient descent to optimize the parameters W and b for a single neuron.
    It follows the lecture pseudo code with some printings to monitor the cost function and its convergence.
    """
    # This version does not have loops on backpropagation because it is for just one neuron
    N = X.shape[0]
    stopping = False
    J_running = 0
    J_running_prev = 0
    iteration = 0

    while not stopping:
        # Randomly select one data point #Sample size = 1
        i_random = np.random.randint(0, high=N)
        y_hat = predict(X[i_random], W, b)
        # Compute cost function for the selected data point
        J_current = -(y[i_random] * np.log(y_hat) + (1 - y[i_random]) * np.log(1 - y_hat))[0]
        error = y_hat - y[i_random]

        # This backpropagation is naive because it is for just one neuron.
        delta_W = error * X[i_random]
        delta_b = error

        # Update parameters with learning rate alpha
        W -= alpha * delta_W
        b -= alpha * delta_b

        # Check stopping criteria
        iteration += 1
        J_running += J_current
        if iteration >= max_iters:
            stopping = True
            print("Stopping by reaching maximum iterations:", max_iters)
        # Check cost function every N iterations--1 epoch
        if iteration % N == 0:
            J_avg = J_running / N
            if iteration % (50*N) == 0:
```

```

        print("Epoch:", int(iteration/N), "    Cost:", round(J_avg,6),
              " Previous Cost:", round(J_running_prev,6),
              " Difference:", round(abs(J_avg - J_running_prev),6)
              )
    # Check for convergence using the difference in average cost over epoch because it was more stable than comparing t
    if abs(J_avg - J_running_prev) < threshold:
        stopping = True
        print("Stopping by convergence at epoch:", int(iteration/N))
        print("Difference in cost function:", abs(J_avg - J_running_prev))

    J_running_prev = J_avg
    J_running = 0

    return W, b

```

Task 2 Implement Logistic Regression

Requirements

- Read both files
- Split train(70), test(15), validation(15)
- Train logistic regressor and check performance
- Scatter plot of results

```

# Reading data function
def read_data(file_path):
    df = pd.read_csv(file_path)
    y = df['Class'].values # Labels
    X = df.drop('Class', axis=1).values # Features
    return X, y

```

```

def split_data(X, y, train_ratio=0.7, test_ratio=0.15, validation_ratio=0.15, random_state=1):
    """This function splits the data into training, testing, and validation sets based on the specified proortion.
    It uses the train_test_split function twice from scikit-learn to perform the splits into 3 sets."""
    # https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.train_test_split.html
    X_train, X_test_pre, y_train, y_test_pre = sklearn.model_selection.train_test_split(
        X, y, train_size=train_ratio, random_state=random_state)

    X_test, X_val, y_test, y_val = sklearn.model_selection.train_test_split(
        X_test_pre, y_test_pre, train_size=test_ratio/(test_ratio + validation_ratio), random_state=random_state)

    return X_train, y_train, X_test, y_test, X_val, y_val

```

Blobs 600 dataset

```

file_path = 'Dataset/blobs600.csv'
X, y = read_data(file_path)
# Split data into training, testing, and validation sets
X_train, y_train, X_test, y_test, X_val, y_val = split_data(X, y)
# Initialize weights and bias
np.random.seed(3301)
W = np.random.normal(0,0.01,X.shape[1])
b = np.random.normal(0,0.01,1)

# Training the model using Stochastic Gradient Descent
W_trained, b_trained = stochastic_gradient_descent(X_train, y_train, W, b, alpha=1e-3, max_iters=100000, threshold=1e-4)

Epoch: 50    Cost: 0.034931 Previous Cost: 0.032443 Difference: 0.002488
Epoch: 100   Cost: 0.028842 Previous Cost: 0.024063 Difference: 0.004779
Stopping by convergence at epoch: 135
Difference in cost function: 7.126158901130802e-05

```

```

# Decision boundary to classify the predicted probability into binary classes
treshold = 0.5

y_hat_train = predict(X_train, W_trained, b_trained)
y_pred_train = (y_hat_train >= treshold).astype(int)

y_hat_test = predict(X_test, W_trained, b_trained)

```

```

y_pred_test = (y_hat_test >= threshold).astype(int)

y_hat_val = predict(X_val, W_trained, b_trained)
y_pred_val = (y_hat_val >= threshold).astype(int)

# Calculate accuracy for each set using np.mean
accuracy_train = np.mean(y_pred_train == y_train)
accuracy_test = np.mean(y_pred_test == y_test)
accuracy_val = np.mean(y_pred_val == y_val)
print("Train Accuracy:", round(accuracy_train*100, 3), "%")
print("Test Accuracy:", round(accuracy_test*100, 3), "%")
print("Validation Accuracy:", round(accuracy_val*100, 3), "%")

```

```

Train Accuracy: 99.762 %
Test Accuracy: 98.889 %
Validation Accuracy: 98.889 %

```

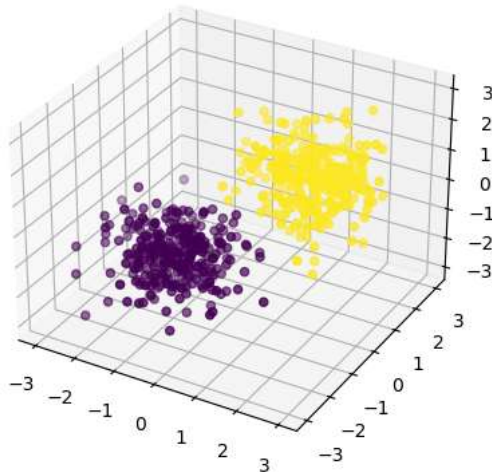
```

# Scatter plot of test results
# https://stackoverflow.com/questions/12414619/creating-a-3d-plot-from-a-3d-numpy-array
X_all = np.vstack((X_train, X_test, X_val))
y_pred_all = np.hstack((y_pred_train, y_pred_test, y_pred_val))

fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')
ax.scatter(X_all[:, 0], X_all[:, 1], X_all[:, 2], c=y_pred_all)

```

```
<mpl_toolkits.mplot3d.art3d.Path3DCollection at 0xac8c630>
```



Accuracy for blobs dataset is high since it was linearly separable.

It reaches convergence after **135 epochs** with a **learning rate = 1e-3** and a **threshold = 1e-4**

Plotting the results was very helpful to understand the dataset and its attributes

✓ Circles 500 dataset

```

file_path = 'Dataset/circles500.csv'
X, y = read_data(file_path)
# Split data into training, testing, and validation sets
X_train, y_train, X_test, y_test, X_val, y_val = split_data(X, y)
# Initialize weights and bias
np.random.seed(3301)
W = np.random.normal(0,0.01,X.shape[1])
b = np.random.normal(0,0.01,1)

# Training the model using Stochastic Gradient Descent
W_trained, b_trained = stochastic_gradient_descent(X_train, y, W, b, alpha=1e-4, max_iters=100000, threshold=1e-6)

```

```

Epoch: 50    Cost: 0.691728 Previous Cost: 0.691519 Difference: 0.000209
Epoch: 100   Cost: 0.692641 Previous Cost: 0.689862 Difference: 0.002779
Epoch: 150   Cost: 0.68902  Previous Cost: 0.687235 Difference: 0.001785
Epoch: 200   Cost: 0.689066 Previous Cost: 0.6859  Difference: 0.003166
Epoch: 250   Cost: 0.687737 Previous Cost: 0.689558 Difference: 0.001821
Stopping by reaching maximum iterations: 100000

```

```
# Decision boundary to classify the predicted probability into binary classes
threshold = 0.5

y_hat_train = predict(X_train, W_trained, b_trained)
y_pred_train = (y_hat_train >= threshold).astype(int)

y_hat_test = predict(X_test, W_trained, b_trained)
y_pred_test = (y_hat_test >= threshold).astype(int)

y_hat_val = predict(X_val, W_trained, b_trained)
y_pred_val = (y_hat_val >= threshold).astype(int)

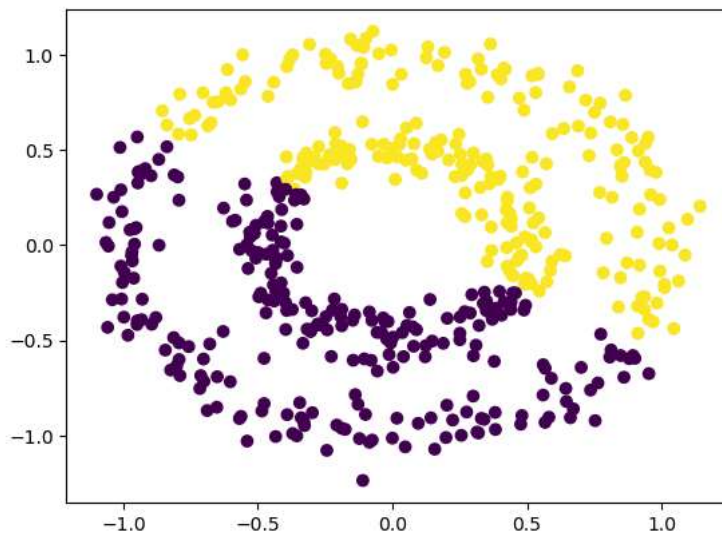
# Calculate accuracy for each set using np.mean
accuracy_train = np.mean(y_pred_train == y_train)
accuracy_test = np.mean(y_pred_test == y_test)
accuracy_val = np.mean(y_pred_val == y_val)
print("Train Accuracy:", round(accuracy_train*100, 3), "%")
print("Test Accuracy:", round(accuracy_test*100, 3), "%")
print("Validation Accuracy:", round(accuracy_val*100, 3), "%")
```

```
Train Accuracy: 48.286 %
Test Accuracy: 54.667 %
Validation Accuracy: 48.0 %
```

```
# Scatter plot of test results
# https://stackoverflow.com/questions/12414619/creating-a-3d-plot-from-a-3d-numpy-array
X_all = np.vstack((X_train, X_test, X_val))
y_pred_all = np.hstack((y_pred_train, y_pred_test, y_pred_val))

fig = plt.figure()
ax = fig.add_subplot()
ax.scatter(X_all[:, 0], X_all[:, 1], c=y_pred_all)
```

```
<matplotlib.collections.PathCollection at 0xaa12f60>
```



Accuracy for circles dataset is low and basically as good as a random choice.

This is because it is not linearly separable and a simple logistic regression cannot approximate to a more complex function.

The plot shows that the neuron trained a linear function.

Task 3: Implement and Test a Shallow Neural Network

Requirements

- 1 hidden layer
- 1 output node
- Support different number of inputs
- Support different number of nodes in the hidden layer

Test

- Test previous datasets with different number of nodes in the hidden layer
- Measure performance

```
def parameter_initialization(X, neurons_hidden_layer):
    """This function initializes parameters (W,b) for a neural network with one hidden layer.
    neurons_hidden_layer is the number of neurons in the hidden layer.
    The weights and bias are initialized with small random values from a normal distribution.
    z and a are initialized with zeros just to create the shape of them for the forward and backward propagation steps."""
    np.random.seed(3301)
    # Weight parameters
    W1 = np.random.normal(0,0.01,size=(neurons_hidden_layer,X.shape[1]))
    W2 = np.random.normal(0,0.01,size=(1,neurons_hidden_layer))
    W = [W1, W2]
    # Bias parameters
    b1 = np.random.normal(0,0.01,size=(neurons_hidden_layer))
    b2 = np.random.normal(0,0.01,size=(1))
    b = [b1, b2]

    # https://numpy.org/devdocs/reference/generated/numpy.zeros.html
    z1 = np.zeros(shape=(neurons_hidden_layer))
    z2 = np.zeros(shape=(1))
    z = [z1, z2]
    a0 = np.zeros(shape=(X.shape[1]))
    a1 = np.zeros(shape=(neurons_hidden_layer))
    a2 = np.zeros(shape=(1))
    a = [a0, a1, a2]

    return W, b, z, a
```

```
# Forward propagation function
def forward_propagation(Xi, W, b, z, a):
    """Forward propagation function for a single input Xi"""
    # X[observation][feature]
    # W[layer][destination_node][source_node]
    # b[layer][destination_node]
    # z[layer][destination_node]
    # a[layer][destination_node]
    a[0] = Xi # Inputs are activations of layer 0
    for layer in range(1, len(W)+1):
        for destination_node in range(1, len(z[layer-1])+1):
            # Linear transformation
            z[layer-1][destination_node-1] = np.dot(a[layer-1], W[layer-1][destination_node-1]) + b[layer-1][destination_node-1]
            # Activation function (Using logistic function)
            a[layer][destination_node-1] = 1 / (1 + np.exp(-z[layer-1][destination_node-1]))

    y_hat = a[layer][destination_node-1] # Output of the last layer is the prediction
    return y_hat, a, z
```

```
# Return only the prediction
def predict_t3(X, W, b, z, a):
    """Prepares forward propagation function for multiple input samples X"""
    y_hats = np.zeros(shape=(X.shape[0]))
    for i, Xi in enumerate(X):
        y_hat, a, z = forward_propagation(Xi, W, b, z, a)
        y_hats[i] = y_hat
    return y_hats
```

```
# Backpropagation and gradient descent function
def gradient_descent_t3(X, y, W, b, a, z, alpha=0.01, max_iters=1000, threshold=1e-4, neurons_hidden_layer=5):
    N = X.shape[0]
    stopping = False
    J_running = 0
    J_running_prev = 0
    iteration = 0
    # Initialize delta parameters for gradient descent
    delta_W1 = np.zeros(shape=(neurons_hidden_layer,X.shape[1]))
    delta_W2 = np.zeros(shape=(1,neurons_hidden_layer))
    delta_W = [delta_W1, delta_W2]

    delta_b1 = np.zeros(shape=(neurons_hidden_layer))
    delta_b2 = np.zeros(shape=(1))
    delta_b = [delta_b1, delta_b2]

    delta_z1 = np.zeros(shape=(neurons_hidden_layer))
```

```

delta_z2 = np.zeros(shape=(1))
delta_z = [delta_z1, delta_z2]

while not stopping:
    # Randomly select one data point #Sample size = 1
    i_random = np.random.randint(0, high=N)
    y_hat, a, z = forward_propagation(X[i_random], W, b, z, a)
    J_current = - (y[i_random] * np.log(y_hat) + (1 - y[i_random]) * np.log(1 - y_hat))

    # Backpropagation step
    error = y_hat - y[i_random]
    for layer in range(len(W), 0, -1):
        # Output layer gradients
        if layer == len(W):
            delta_z[layer-1] = error
            # https://numpy.org/doc/2.3/reference/generated/numpy.reshape.html
            delta_W[layer-1] = error * a[layer-1].reshape(1, -1)
            delta_b[layer-1] = error
        # Hidden layer gradients
        else:
            # Derivative of logistic is f(z) * (1 - f(z)).
            # Therefore, we can use the activation layer instead of calculating the logistic function again
            delta_z[layer-1] = (a[layer] * (1 - a[layer])) * delta_z[layer] * W[layer].reshape(-1)
            # https://numpy.org/doc/stable/reference/generated/numpy.outer.html
            delta_W[layer-1] = np.outer(delta_z[layer-1], a[layer-1])
            delta_b[layer-1] = delta_z[layer-1]

    # Update parameters step
    for layer in range(1, len(W)+1):
        W[layer-1] -= alpha * delta_W[layer-1]
        b[layer-1] -= alpha * delta_b[layer-1]

    # Check stopping criteria
    iteration += 1
    J_running += J_current
    if iteration >= max_iters:
        stopping = True
        print("Stopping by reaching maximum iterations:", max_iters)
    if iteration % N == 0: # 1 Epoch finished
        J_avg = J_running / N
        if iteration % (2000*N) == 0:
            print("Epoch:", int(iteration/N), " Cost:", round(J_avg,6),
                  " Previous Cost:", round(J_running_prev,6),
                  " Difference:", round(abs(J_avg - J_running_prev),6)
                  )
        if abs(J_avg - J_running_prev) < threshold:
            stopping = True
            print("Stopping by convergence at epoch:", int(iteration/N))
            print("Difference in cost function:", abs(J_avg - J_running_prev))
        J_running_prev = J_avg
        J_running = 0

    return W, b, a, z

```

▼ Blobs 600 dataset

```

file_path = 'Dataset/blobs600.csv'
X, y = read_data(file_path)
# Split data into training, testing, and validation sets
X_train, y_train, X_test, y_test, X_val, y_val = split_data(X, y)
# Initialize weights and bias
np.random.seed(3301)
neurons_hidden_layer = 4
W, b, z, a = parameter_initialization(X, neurons_hidden_layer=neurons_hidden_layer)

# Training the model using Stochastic Gradient Descent
W_trained, b_trained, a_trained, z_trained = gradient_descent_t3(X_train, y_train, W, b, a, z,
                                                                  alpha=1e-3, max_iters=800000,
                                                                  threshold=1e-5,
                                                                  neurons_hidden_layer=neurons_hidden_layer)

```

Stopping by convergence at epoch: 1393
 Difference in cost function: 8.380746544547207e-06

```
# Decision boundary to classify the predicted probability into binary classes
threshold = 0.5

y_hat_train = predict_t3(X_train, W_trained, b_trained, z_trained, a_trained)
y_pred_train = (y_hat_train >= threshold).astype(int)

y_hat_test = predict_t3(X_test, W_trained, b_trained, z_trained, a_trained)
y_pred_test = (y_hat_test >= threshold).astype(int)

y_hat_val = predict_t3(X_val, W_trained, b_trained, z_trained, a_trained)
y_pred_val = (y_hat_val >= threshold).astype(int)

# Calculate accuracy for each set using np.mean
accuracy_train = np.mean(y_pred_train == y_train)
accuracy_test = np.mean(y_pred_test == y_test)
accuracy_val = np.mean(y_pred_val == y_val)
print("Train Accuracy:", round(accuracy_train*100, 3), "%")
print("Test Accuracy:", round(accuracy_test*100, 3), "%")
print("Validation Accuracy:", round(accuracy_val*100, 3), "%")
```

```
Train Accuracy: 99.762 %
Test Accuracy: 98.889 %
Validation Accuracy: 100.0 %
```

✓ Circles 500 dataset

```
file_path = 'Dataset/circles500.csv'
X, y = read_data(file_path)
# Split data into training, testing, and validation sets
X_train, y_train, X_test, y_test, X_val, y_val = split_data(X, y)
# Initialize weights and bias
np.random.seed(3301)
neurons_hidden_layer = 5
W, b, z, a = parameter_initialization(X, neurons_hidden_layer=neurons_hidden_layer)

# Training the model using Stochastic Gradient Descent
W_trained, b_trained, a_trained, z_trained = gradient_descent_t3(X_train, y_train, W, b, a, z,
                                                                alpha=1e-2, max_iters=1000000,
                                                                threshold=1e-6,
                                                                neurons_hidden_layer=neurons_hidden_layer)
```

```
Epoch: 2000    Cost: 0.108895 Previous Cost: 0.111182 Difference: 0.002287
Epoch: 4000    Cost: 0.028379 Previous Cost: 0.035217 Difference: 0.006838
Epoch: 6000    Cost: 0.014489 Previous Cost: 0.01403  Difference: 0.000459
Stopping by convergence at epoch: 6902
Difference in cost function: 9.783105273560022e-07
```

```
# Decision boundary to classify the predicted probability into binary classes
threshold = 0.5

y_hat_train = predict_t3(X_train, W_trained, b_trained, z_trained, a_trained)
y_pred_train = (y_hat_train >= threshold).astype(int)

y_hat_test = predict_t3(X_test, W_trained, b_trained, z_trained, a_trained)
y_pred_test = (y_hat_test >= threshold).astype(int)

y_hat_val = predict_t3(X_val, W_trained, b_trained, z_trained, a_trained)
y_pred_val = (y_hat_val >= threshold).astype(int)

# Calculate accuracy for each set using np.mean
accuracy_train = np.mean(y_pred_train == y_train)
accuracy_test = np.mean(y_pred_test == y_test)
accuracy_val = np.mean(y_pred_val == y_val)
print("Train Accuracy:", round(accuracy_train*100, 3), "%")
print("Test Accuracy:", round(accuracy_test*100, 3), "%")
print("Validation Accuracy:", round(accuracy_val*100, 3), "%")
```

```
Train Accuracy: 100.0 %
Test Accuracy: 100.0 %
Validation Accuracy: 100.0 %
```

The neural network now is able to handle linear and non linear data performing better in the circle dataset.

If there are few neurons in the hidden layer it is not able to represent the function so it has poor performance. If there are many neurons

the training and convergence becomes slower

Best configuration in our case was:

learning rate = $1e-2$

neurons in hidden layer ≥ 4

threshold = $1e-6$

✓ Task 4: Tackle a Challenging Task

Requirements

- Read image dataset
- Filter two letters (C,F)
- Reshape images (flatten it)
- 1 hidden layer
- 1 output node
- Support different number of inputs
- Support different number of nodes in the hidden layer

Test

- Check accuracy
- Plot training curve

```
# Backpropagation and gradient descent function
def gradient_descent_t4(X, y, W, b, a, z, alpha=0.01, max_iters=1000, threshold=1e-4, neurons_hidden_layer=5):
    N = X.shape[0]
    stopping = False
    J_running = 0
    J_running_prev = 0
    iteration = 0
    # Initialize delta parameters for gradient descent
    delta_W1 = np.zeros(shape=(neurons_hidden_layer,X.shape[1]))
    delta_W2 = np.zeros(shape=(1,neurons_hidden_layer))
    delta_W = [delta_W1, delta_W2]

    delta_b1 = np.zeros(shape=(neurons_hidden_layer))
    delta_b2 = np.zeros(shape=(1))
    delta_b = [delta_b1, delta_b2]

    delta_z1 = np.zeros(shape=(neurons_hidden_layer))
    delta_z2 = np.zeros(shape=(1))
    delta_z = [delta_z1, delta_z2]

    J_diff_array = []

    while not stopping:
        # Randomly select one data point #Sample size = 1
        i_random = np.random.randint(0, high=N)
        y_hat, a, z = forward_propagation(X[i_random], W, b, z, a)
        J_current = - (y[i_random] * np.log(y_hat) + (1 - y[i_random]) * np.log(1 - y_hat))

        # Backpropagation step
        error = y_hat - y[i_random]
        for layer in range(len(W), 0, -1):
            # Output layer gradients
            if layer == len(W):
                delta_z[layer-1] = error
                # https://numpy.org/doc/2.3/reference/generated/numpy.reshape.html
                delta_W[layer-1] = error * a[layer-1].reshape(1, -1)
                delta_b[layer-1] = error
            # Hidden layer gradients
            else:
                # Derivative of logistic is f(z) * (1 - f(z)). Therefore, we can use the activation layer instead of calculating it
                delta_z[layer-1] = (a[layer] * (1 - a[layer])) * delta_z[layer] * W[layer].reshape(-1)
                # https://numpy.org/doc/stable/reference/generated/numpy.outer.html
                delta_W[layer-1] = np.outer(delta_z[layer-1], a[layer-1])
                delta_b[layer-1] = delta_z[layer-1]

        # Update parameters step
        for layer in range(1, len(W)+1):
```

```

        W[layer-1] -= alpha * delta_W[layer-1]
        b[layer-1] -= alpha * delta_b[layer-1]

# Check stopping criteria
iteration += 1
J_running += J_current
if iteration >= max_iters:
    stopping = True
    print("Stopping by reaching maximum iterations:", max_iters)
    print("Difference in cost function:", abs(J_running / N - J_running_prev))
if iteration % N == 0: # 1 Epoch finished
    J_avg = J_running / N
    if iteration % (2000*N) == 0:
        print("Epoch:", int(iteration/N), "    Cost:", round(J_avg,6),
              " Previous Cost:", round(J_running_prev,6),
              " Difference:", round(abs(J_avg - J_running_prev),6)
              )
        J_diff_array.append(abs(J_avg - J_running_prev))
    if abs(J_avg - J_running_prev) < threshold:
        stopping = True
        print("Stopping by convergence at epoch:", int(iteration/N))
        print("Difference in cost function:", abs(J_avg - J_running_prev))
    J_running_prev = J_avg
    J_running = 0

plt.plot(J_diff_array, label='Difference in Cost')
plt.xlabel('Epochs (Every N iterations)')
plt.ylabel('Cost')
plt.title('Cost Function over Epochs')
plt.legend()
plt.show()

return W, b, a, z

```

```

# Reading images
data = np.load("Dataset/emnist_letters_85800.npz")
x_data = data["x"]
y_data = data["y"]

```

```

# Extract just two classes from the dataset
c1 = 3 # C
c2 = 6 # F

```

```

mask = (y_data == c1) | (y_data == c2)
x_binary = x_data[mask]
y_binary = y_data[mask]
# Now change labels to 0 and 1
y_binary = (y_binary == c2).astype(int)

```

```

# Do some data checks ...

```

```

# Check how many different classes we have: should be 26
n_classes = len(np.unique(y_data))
print(f"The number of unique classes is {n_classes} (should be 26).")
# check that images are scaled: min should be 0, max should be 1
img = x_data[0]
print(f"For a single image, the min value is {img.min()} and the max is {img.max()} (should be 0.0 and 1.0).")
# Check the shape of the two classes
print(f"Shape of x_data is {x_data.shape}, shape of y_data is {y_data.shape} (should have 85,800 cases and x should be 28x28).")

```

The number of unique classes is 26 (should be 26).
 For a single image, the min value is 0.0 and the max is 1.0 (should be 0.0 and 1.0).
 Shape of x_data is (85800, 28, 28, 1), shape of y_data is (85800,) (should have 85,800 cases and x should be 28x28).

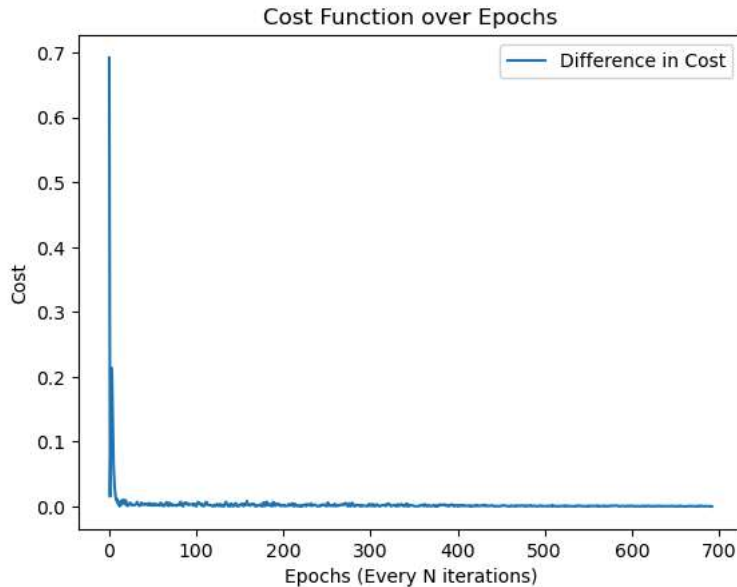
```

# Flatten the images
x_flat = x_binary.reshape(x_binary.shape[0], -1)
# Split data into training, testing, and validation sets
X_train, y_train, X_test, y_test, X_val, y_val = split_data(x_flat, y_binary)
# Initialize weights and bias
np.random.seed(3301)
neurons_hidden_layer = 8
W, b, z, a = parameter_initialization(x_flat, neurons_hidden_layer=neurons_hidden_layer)
# Training the model using Stochastic Gradient Descent
W_trained, b_trained, a_trained, z_trained = gradient_descent_t4(X=X_train, y=y_train,
                                                                W=W, b=b, a=a, z=z,

```

```
alpha=1e-3, max_iters=10000000,
threshold=1e-6,
neurons_hidden_layer=neurons_hidden_layer)
```

Stopping by convergence at epoch: 693
 Difference in cost function: 3.8377293383549255e-07



```
treshold = 0.5
y_hat_train= predict_t3(X_train, W_trained, b_trained, z_trained, a_trained)
y_pred_train = (y_hat_train >= treshold).astype(int)

y_hat_test= predict_t3(X_test, W_trained, b_trained, z_trained, a_trained)
y_pred_test = (y_hat_test >= treshold).astype(int)

y_hat_val= predict_t3(X_val, W_trained, b_trained, z_trained, a_trained)
y_pred_val = (y_hat_val >= treshold).astype(int)

accuracy_train = np.mean(y_pred_train == y_train)
accuracy_test = np.mean(y_pred_test == y_test)
accuracy_val = np.mean(y_pred_val == y_val)
print("Train Accuracy:", round(accuracy_train*100, 3), "%")
print("Test Accuracy:", round(accuracy_test*100, 3), "%")
print("Validation Accuracy:", round(accuracy_val*100, 3), "%")
```

```
Train Accuracy: 100.0 %
Test Accuracy: 99.091 %
Validation Accuracy: 98.788 %
```

The neural network is able to effectively classify the 2 letters.

It could be because c and f have clear different contours that are easy to detect with a small network

Good results (convergence) are find with:

learning rate = 1e-3

threshold = 1e-6

neurons in hidden layer >= 6

✓ Task 5: Deep Learning enhancements

Requirements

- Support any number of layers - Done
- Support any number of neurons in each layer - Done
- Implement method to avoid overfitting - Done

Test

- Test different layers and with or without overfitting method - Done

- Check accuracy - Done
- Plot training curve - Done

```
# Reading images
# data = np.load("data/emnist_letters_85800.npz")
data = np.load("Dataset/emnist_letters_85800.npz")
x_data = data["x"]
y_data = data["y"]

# Extract just two classes from the dataset
c1 = 3 # C
c2 = 6 # F

mask = (y_data == c1) | (y_data == c2)
x_binary = x_data[mask]
y_binary = y_data[mask]
# Now change labels to 0 and 1
y_binary = (y_binary == c2).astype(int)
```

```
# hidden_neurons_per_layer = [3,4,5,6,7,8,9,10]
# each position in the array is the hidden layer and the value is the number of neurons in that layer
def robust_parameter_initialization(X, hidden_neurons_per_layer): #3,4,5,6,7,8,9,10]
    W = []
    b = []
    z = []
    a = []
    np.random.seed(3301)

    a.append(np.zeros(shape=(X.shape[1]))) # a0
    for i, num_neurons in enumerate(hidden_neurons_per_layer):
        if i == 0:
            prev_layer_neurons = X.shape[1]
        else:
            prev_layer_neurons = hidden_neurons_per_layer[i-1]
        # Create weight matrix for the current layer
        W_aux = np.random.normal(0,0.01,size=(num_neurons,prev_layer_neurons))
        W.append(W_aux)
        # Create bias vector for the current layer
        b_aux = np.random.normal(0,0.01,size=(num_neurons))
        b.append(b_aux)
        # Create z vector
        z_aux = np.zeros(shape=(num_neurons))
        z.append(z_aux)
        # Create activation vector a
        a_aux = np.zeros(shape=(num_neurons))
        a.append(a_aux)
        # print(i, num_neurons)

    W_aux_l1 = np.random.normal(0,0.01,size=(1,num_neurons))
    W.append(W_aux_l1)
    b_aux_l1 = np.random.normal(0,0.01,size=(1))
    b.append(b_aux_l1)
    z_aux_l1 = np.zeros(shape=(1))
    z.append(z_aux_l1)
    a_aux_l1 = np.zeros(shape=(1))
    a.append(a_aux_l1)

    return W, b, z, a
```

```
# Forward propagation function
def forward_prop(Xi, W, b, z, a):
    a[0] = Xi # Inputs are activations of layer 0
    for layer in range(len(W)):
        # Linear transformation
        z[layer] = np.dot(W[layer], a[layer]) + b[layer]
        # Activation function (Using logistic function)
        a[layer + 1] = 1 / (1 + np.exp(-z[layer]))

    y_hat = a[len(a) - 1][0] # Output of the last layer is the prediction
    return y_hat, a, z
```

```
def predict_DNN(X, W, b, z, a):
    y_hat_arr = np.zeros(shape=(X.shape[0]))
    for i, Xi in enumerate(X):
```

```

        y_hat, a, z = forward_prop(Xi, W, b, z, a)
        y_hat_arr[i] = y_hat
    return y_hat_arr

```

```

def gradient_descent_l2(X, y, W, b, a, z, alpha, max_iters, threshold, penlambda, hidden_neurons_per_layer):
    N = X.shape[0]
    stopping = False
    J_running = 0
    J_running_prev = 0
    l2 = 0
    J_diff_array = []

    delta_W = []
    delta_b = []
    delta_z = []

    # Initialize delta parameters for hidden layers
    for i, num_neurons in enumerate(hidden_neurons_per_layer):
        if i == 0:
            prev_layer_neuron = X.shape[1]
        else:
            prev_layer_neuron = hidden_neurons_per_layer[i-1]

        delta_W.append(np.zeros(shape=(num_neurons, prev_layer_neuron)))
        delta_b.append(np.zeros(shape=(num_neurons)))
        delta_z.append(np.zeros(shape=(num_neurons)))

    # Initialize delta for output layer
    delta_W.append(np.zeros(shape=(1, hidden_neurons_per_layer[-1])))
    delta_b.append(np.zeros(shape=(1)))
    delta_z.append(np.zeros(shape=(1)))

    iteration = 0

    while not stopping and iteration < max_iters:
        i_random = np.random.randint(0, high=N)

        a_copy = [arr.copy() for arr in a]
        z_copy = [arr.copy() for arr in z]

        y_hat, a_copy, z_copy = forward_prop(X[i_random], W, b, z_copy, a_copy)

        l2 = 0
        for w_arr in W:
            l2 += np.sum(w_arr ** 2)

        J_current = (- (y[i_random] * np.log(y_hat) + (1 - y[i_random]) * np.log(1 - y_hat))) + ((penlambda / (2 * N)) * l2)

        # Backpropagation step
        error = y_hat - y[i_random]

        delta_z_copy = [np.zeros_like(dz) for dz in z_copy]
        delta_W_copy = [np.zeros_like(dw) for dw in W]
        delta_b_copy = [np.zeros_like(db) for db in b]

        for layer in range(len(W)-1, -1, -1):
            # Output layer gradients
            if layer == len(W)-1:
                delta_z_copy[layer] = error
                # https://numpy.org/doc/2.3/reference/generated/numpy.reshape.html
                delta_W_copy[layer] = error * a_copy[layer].reshape(1, -1) + ((penlambda / N) * W[layer])
                delta_b_copy[layer] = error
            else:
                # Hidden layer gradients
                # Derivative of logistic is f(z) * (1 - f(z)). Therefore, we can use the activation layer instead of calculating
                p1 = a_copy[layer+1] * (1 - a_copy[layer+1])
                p2 = np.dot(W[layer+1].T, delta_z_copy[layer+1]).flatten() # flatten needed cause shapes are different
                delta_z_copy[layer] = p1 * p2
                # https://numpy.org/doc/stable/reference/generated/numpy.outer.html
                delta_W_copy[layer] = np.outer(delta_z_copy[layer], a_copy[layer]) + ((penlambda / N) * W[layer])
                delta_b_copy[layer] = delta_z_copy[layer]

        # Update parameters step
        for layer in range(len(W)):

```

```

        W[layer] -= alpha * delta_W_copy[layer]
        b[layer] -= alpha * delta_b_copy[layer]

# Check stopping criteria
iteration += 1
J_running += J_current
if iteration >= max_iters:
    stopping = True
    print("Stopping by reaching maximum iterations:", max_iters)
    print("Difference in cost function:", abs(J_running / N - J_running_prev))
if iteration % N == 0: # 1 Epoch finished
    J_avg = J_running / N
    if iteration % (10*N) == 0:
        print("Epoch:", int(iteration/N), "    Cost:", round(J_avg,6),
              "    Previous Cost:", round(J_running_prev,6),
              "    Difference:", round(abs(J_avg - J_running_prev),6)
              )
        J_diff_array.append(abs(J_avg - J_running_prev))
    if abs(J_avg - J_running_prev) < threshold:
        stopping = True
        print("Stopping by convergence at epoch:", int(iteration/N))
        print("Difference in cost function:", abs(J_avg - J_running_prev))
    J_running_prev = J_avg
    J_running = 0

plt.plot(J_diff_array, label='Difference in Cost')
plt.xlabel('Epochs (Every N iterations)')
plt.ylabel('Cost')
plt.title('Cost Function over Epochs')
plt.legend()
plt.show()

return W, b, a_copy, z_copy

```

larger alpha is needed as fine-tuning the parameters showed smaller performed worse
only 3 hidden layers showed very good performance

learning rate alpha=1e-1

penalty lambda penlambda=1e-2

hidden_neurons_per_layer=[3,4,5] -> number of layers = 3

```

x_flat = x_binary.reshape(x_binary.shape[0], -1)
# Split data into training, testing, and validation sets
X_train, y_train, X_test, y_test, X_val, y_val = split_data(x_flat, y_binary)
# Initialize weights and bias
np.random.seed(3301)
hidden_neurons_per_layer = [3,4,5]
W, b, z, a = robust_parameter_initialization(X_train, hidden_neurons_per_layer=hidden_neurons_per_layer)

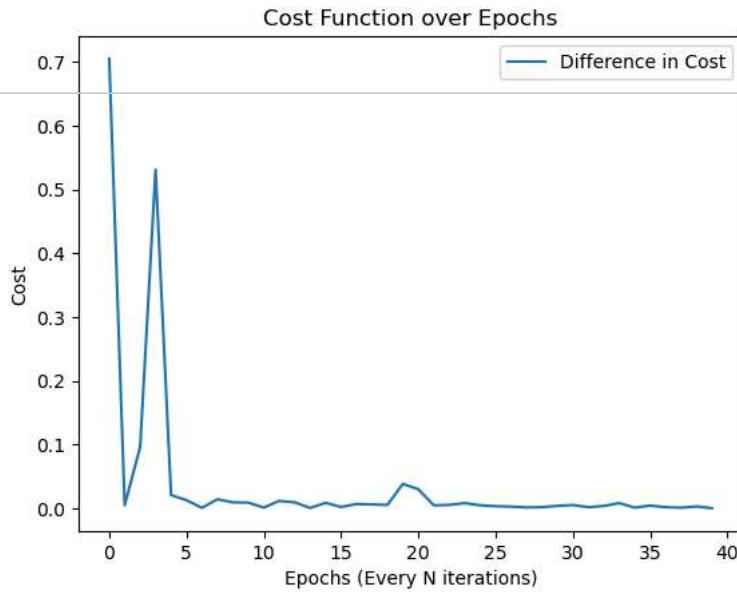
W_trained, b_trained, a_trained, z_trained = gradient_descent_l2(X_train, y_train,
                                                                    W, b, a, z, alpha=1e-1,
                                                                    max_iters=1e6, threshold=1e-4,
                                                                    penlambda=1e-2,
                                                                    hidden_neurons_per_layer=hidden_neurons_per_layer)

```

```

Epoch: 10    Cost: 0.025109 Previous Cost: 0.016221 Difference: 0.008888
Epoch: 20    Cost: 0.046339 Previous Cost: 0.008129 Difference: 0.03821
Epoch: 30    Cost: 0.017057 Previous Cost: 0.013291 Difference: 0.003766
Epoch: 40    Cost: 0.002506 Previous Cost: 0.002526 Difference: 2e-05
Stopping by convergence at epoch: 40
Difference in cost function: 1.9877280302096813e-05

```



```

threshold = 0.01
y_hat_train= predict_DNN(X_train, W_trained, b_trained, z_trained, a_trained)
y_pred_train = (y_hat_train >= threshold).astype(int)

y_hat_test= predict_DNN(X_test, W_trained, b_trained, z_trained, a_trained)
y_pred_test = (y_hat_test >= threshold).astype(int)

y_hat_val= predict_DNN(X_val, W_trained, b_trained, z_trained, a_trained)
y_pred_val = (y_hat_val >= threshold).astype(int)

```

`y_hat_train[:4]` gives values array([3.58861190e-04, 9.99754338e-01, 3.57660997e-04, 3.12796387e-04])
 so the threshold needs to be between 1e-4 and 1e-1 and we chose 1e-2

```

# Calculate accuracy for each set using np.mean
accuracy_train = np.mean(y_pred_train == y_train)
accuracy_test = np.mean(y_pred_test == y_test)
accuracy_val = np.mean(y_pred_val == y_val)
print("Train Accuracy:", round(accuracy_train*100, 3), "%")
print("Test Accuracy:", round(accuracy_test*100, 3), "%")
print("Validation Accuracy:", round(accuracy_val*100, 3), "%")

```

```

Train Accuracy: 99.199 %
Test Accuracy: 99.091 %
Validation Accuracy: 99.091 %

```